Mth401 Quiz No:02



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Question # 4 of 10 (Start time: 02:35:28 PM, 17 June 2021) Total Mar

The differential operator D transforms a _____ function into another function.

Select the correct option

- discontinuous
- integrable
- differentiable
- continuous

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Question # 6 of 10 (Start time: 02:39:32 PM, 17 June 2021)

A set of functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is said to be _____ on an interval I if there exist constants

$$c_1, c_2, c_3, \ldots\ldots, c_n$$

not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x), \ldots\ldots, c_n f_n(x) = 0, \quad \forall x \in I$$

Select the correct option

- linear independent
- linear dependent

P-100 ✓

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13.5 Linear Dependence

A set of functions

$$\{f_1(x), f_2(x), \ldots, f_n(x)\}$$

is said to be ***linearly dependent*** on an interval I if $\exists$ constants $c_1, c_2, \ldots, c_n$ not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0, \quad \forall\, x \in I$$

13.6 Linear Independence

A set of functions $\{f_1(x), f_2(x), \ldots, f_n(x)\}$ is said to be linearly independent on an interval I if $c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0, \quad \forall\, x \in I$,only when $c_1 = c_2 = \cdots = c_n = 0.$

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Question # 7 of 10 (Start time: 02:41:46 PM, 17 June 2021)

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

Select the correct option

- $m^2 - 3 = 0$
- $m^3 + 2m^2 - 3 = 0$ ✓
- $m^3 + 2m^2 = 0$
- $\left(m^3 + 2m^2 - 3\right) e^{mx} = 0$

Click to Save Answer & Move

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MTH401:Quiz NO.02

Quiz Start Time: 02:30 PM, 17 June 2021

Question # 9 of 10 (Start time: 02:43:31 PM, 17 June 2021)

Total Marks: 1

The displacement measured below the equilibrium position are _____.

Select the correct option

- negative
- positive

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 02:44:44 PM, 17 June 2021) Total Marks: 1

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- distributive
- inverse
- identical
- commute



Click to Save / Answer & Move to Next Question



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Question # 1 of 10 (Start time: 02:44:57 PM, 17 June 2021)

If

$$a_n(x) = 0$$

in the differential equation

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \ldots\ldots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

for some

$$x \in I$$

Select the correct option

○	none of these ✓
○	unique
○	exist and unique
○	exist

Click to Save Answer & Move

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Question # 3 of 10 (Start time: 02:47:58 PM, 17 June 2021) **Total Marks:** 1

For 2nd order linear non-homogeneous differential equation

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

, which of the following is NOT the pair of boundary condition _____.

Select the correct option

Reload Math Equations

- $y'(a) = y'_0, \quad y'(b) = y'_1$
- $y'(a) = y'_0, \quad y(b) = y_1$
- $y'(a) = y'_0, \quad y(b) = y_1$
- $y'(a) = y'_0, \quad y'(b) = y'_0$

Click to Save Answer & Move to Next Question

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MTH401:Quiz NO.02 Quiz Start Time: (

Question # 8 of 10 (Start time: 02:54:03 PM, 17 June 2021)

What is annihilator operator of the function

$$g(x) = 8e^{3x}$$

?

Select the correct option

○	$(D+3)^2$
○	$(D-3)^2$
○	$(D+3)$
○	$(D-3)$

Click to Save Answer & Move

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stion # 6 of 10 (Start time: 02:51:55 PM, 17 June 2021) Total M

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.

If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

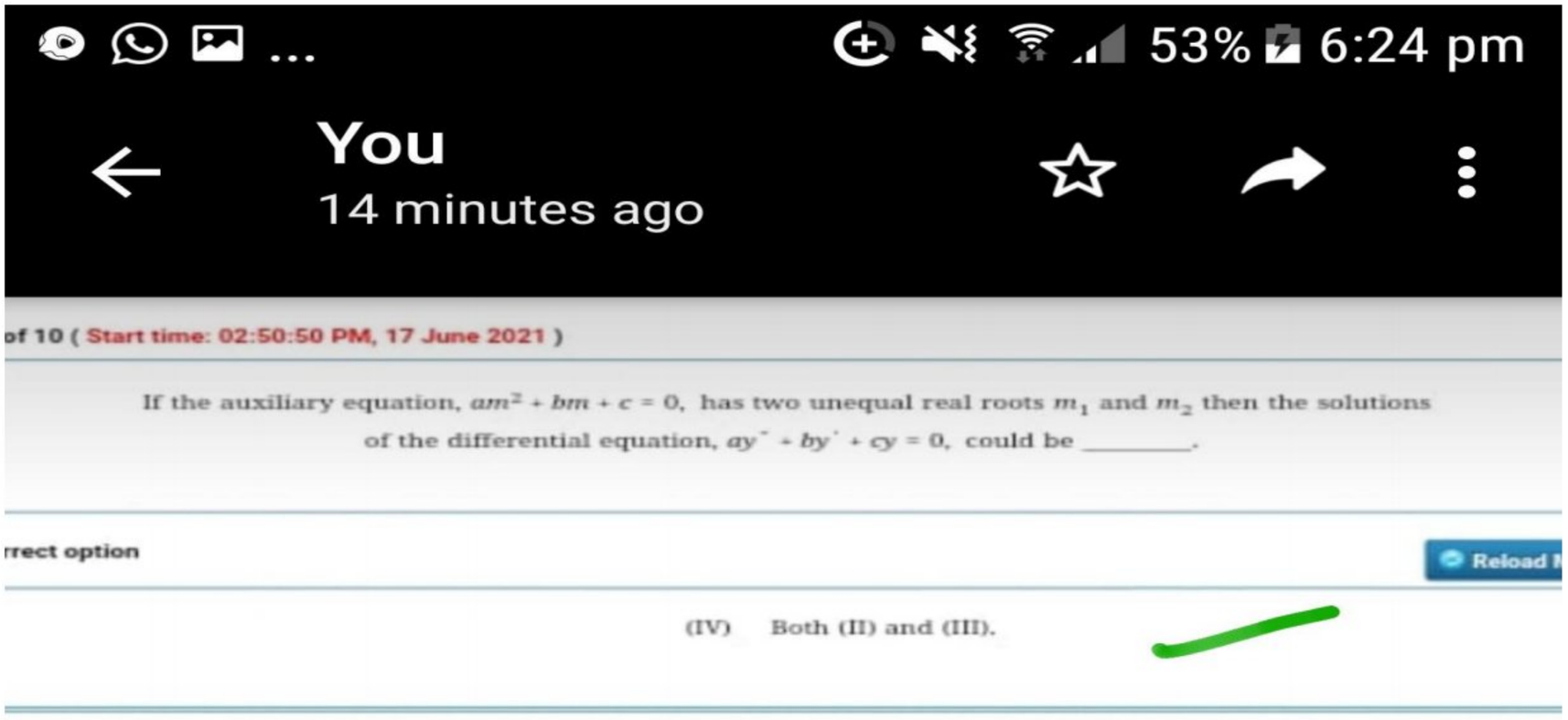
ect the correct option

Reload Math Equ

- $(Ax + B)e^{5x}$
- $(Ax^2 + Bx + C)e^{5x}$
- $Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
- Ae^{5x}

Click to Save Answer & Move to Next Question

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Question # 4 of 10 (Start time: 02:49:28 PM, 17 June 2021)

If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

- Its annihilator operator is (D+5)
- Its annihilator operator is $(D+5)^2$
- Its annihilator operator is (D-5)
- Its annihilator operator is $(D-5)^2$

Click to Save Answer & Mo

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of 10 (Start time: 02:50:50 PM, 17 June 2021)

If the auxiliary equation, $am^2 + bm + c = 0$, has two unequal real roots m_1 and m_2 then the solutions of the differential equation, $ay'' + by' + cy = 0$, could be _____.

rrect option

Reload

(IV) Both (II) and (III).

(III) $y_2 = e^{m_2 x}$

(II) $y_1 = e^{m_1 x}$

(I) $y = c\,e^{mx}$

Click to Save Answer & Move to Next Q

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Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.

If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

ect the correct option

Reload Math

- $(Ax + B)e^{5x}$
- $(Ax^2 + Bx + C)e^{5x}$
- $Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
- Ae^{5x}

Click to Save Answer & Move to Next Quest

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MTH401:Quiz NO.02

Quiz Start Time: 02:44 PM, 17 June 2021

Question # 7 of 10 (Start time: 02:53:05 PM, 17 June 2021)

Total Marks: 1

A function satisfying the differential equation on the interval I whose graph passes through (x_0, y_0) such that the slope of the curve at the point is the number y'_0 is called solution of the _____ value problem.

Select the correct option

Reload Math Equations

- initial
- boundary

p_95

Question # 10 of 10 (Start time: 02:56:59 PM, 17 June 2021) To

If

$$y = e^{(1/2)x}(c_1 \cos\frac{\sqrt{3}}{2}x + c_2 \sin\frac{\sqrt{3}}{2}x)$$

is the complementary solution of

$$y'' - y' + y = 2\sin 3x$$

, then the general form of its particular solution is

y_p

Select the correct option

Reload Mat

- $3\cos 3x + 3\sin 3x$
- $3A\cos 3x + 3A\sin 3x$
- $A\cos 3x + B\sin 3x$
- $3Asin3x + 3B\cos 3x$

Click to Save Answer & Move to Next Ques

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Question # 5 of 10 (Start time: 02:37:17 PM, 17 June 2021) Total Marks: 1

Wronskian

$$W(x, 2x)$$

=_____

Select the correct option

Reload Math Equations

- -1
- None of these
- 1
- 0

Click to Save Answer & Move to Next Question

$$W = \begin{vmatrix} x & 2x \\ 1 & 2 \end{vmatrix}$$

$$= 2x - 2x$$

$$= 0$$

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MTH401:Quiz NO.02

Quiz Start Time: 02:30 PM, 17 June 2021

Question # 3 of 10 (Start time: 02:33:43 PM, 17 June 2021)

Total Marks: 1

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

Reload Math Equations

- $-3x$
- 1
- $3x$
- 0

Click to Save Answer & Move to Next Question

$u \quad -3u$

$1 \quad -3$

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MC210200757: ABBAS RAZA

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MTH401:Quiz NO.02

Quiz Start Time: 10:11 AM, 17 Jun

Question # 7 of 10 (Start time: 10:16:55 AM, 17 June 2021)

Total M

The velocity of a particle moving with simple harmonic motion is ---------- at the mean position.

Select the correct option

- None of these
- Maximum
- Zero
- Minimum

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 03:55:45 PM, 17 June 2021)

For 2nd order linear non-homogeneous differential equation

$$a_2(x)\frac{d^2y}{dx^2} + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

, which of the following is NOT the pair of boundary condition _____.

Select the correct option

○	$y'(a) = y_0', \qquad y'(b) = y_0'$
○	$y'(a) = y_0', \qquad y'(b) = y_1'$
○	$y'(a) = y_0', \quad y(b) = y_1$
○	$y'(a) = y_0', \qquad y(b) = y_1$

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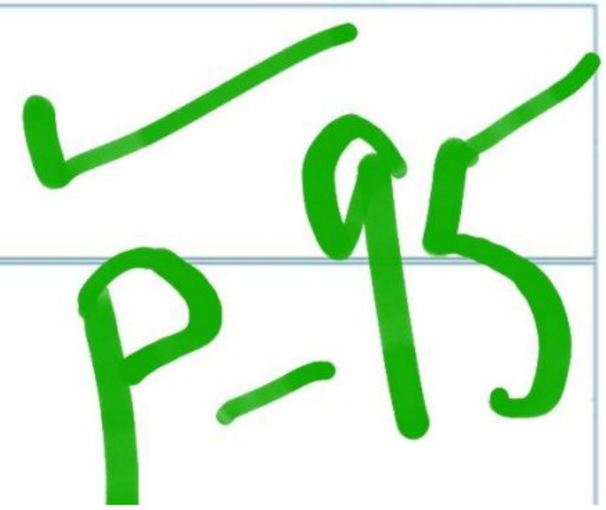
Question # 7 of 10 (Start time: 02:53:05 PM, 17 June 2021) Total Marks: 1

A function satisfying the differential equation on the interval I whose graph passes through (x_0, y_0) such that the slope of the curve at the point is the number y'_0 is called solution of the _____ value problem.

Select the correct option

Reload Math Equations

- initial
- boundary



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Question # 1 of 10 (Start time: 02:44:57 PM, 17 June 2021) Total Marks:

If

$$a_n(x) = 0$$

in the differential equation

$$a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \ldots\ldots + a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

for some

$$x \in I$$

Select the correct option

Reload Math Equations

- none of these
- unique
- exist and unique
- exist

Click to Save Answer & Move to Next Question

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Question # 5 of 10 [Start time: 03:07:19 PM, 17 June 2021] Total Marks: 1

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option

- complementary
- exact
- general
- particular

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Question # 4 of 10 (Start time: 02:32:34 PM, 17 June 2021)

The gravitational force exerted by the earth on a body of mass m is called --------- of the body.

Select the correct option

- weight
- Force

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2:31:40 PM, 17 June 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____.

$A \cos 4x + B \sin 4x$



$Ax^2 + Bx + C$

$(A + B)(\cos 4x + \sin 4x)$

Ae^{4x}

Click to Save Answer &

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Quiz List | Quiz | For a second order | Variation of Parameter | Google
quiz.vu.edu.pk/QuizQuestion.aspx?ver=19a94f81-c83d-4074-8375-1acf169a7e0e
Apps CMS POLCOM SPFC Tenants Registratio... Gmail Tenant Form.pdf Hotel Eye Token IP Slides Display

BC170405370: AYESHA GHAFOOR

MTH401:Quiz NO.02

Question # 2 of 10 (Start time: 02:29:49 PM, 17 June 2021)

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 2$, $W_1 = \dfrac{e^{-x}}{x}$, then $u'_1 =$ _____.

Select the correct option

○	$-\dfrac{e^{-x}}{2x}$
○	$2x\,e^{-x}$
○	$\dfrac{e^{-x}}{2x}$
○	$\dfrac{e^{-x}}{x}$



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Question # 1 of 10 (Start time: 02:27:54 PM, 17 June 2021)

In the method of variations of parameter, the derivative of the function, $u_2' =$ _____.

Select the correct option

○	$W \pm W_2$
○	$\frac{W}{W_2}$
○	$\frac{W_2}{W}$
○	$W.\ W_2$

✓ P-170

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$$e^{\alpha x}, xe^{\alpha x}, x^2e^{\alpha x}, \ldots, x^{n-1}e^{\alpha x}.$$

Therefore, the general solution of the differential equation is

$$y = c_1e^{\alpha x} + c_2xe^{\alpha x} + c_3x^2e^{\alpha x} + \cdots + c_nx^{n-1}e^{\alpha x}$$

So that the differential operator $(D-\alpha)^n$

annihilates each of the functions $e^{\alpha x}, xe^{\alpha x}, x^2e^{\alpha x}, \ldots, x^{n-1}e^{\alpha x}$

Hence, as a consequence of the fact that the differentiation can be performed term by term, the differential operator $(D-\alpha)^n$

annihilates the function $y = c_1e^{\alpha x} + c_2xe^{\alpha x} + c_3x^2e^{\alpha x} + \cdots + c_nx^{n-1}e^{\alpha x}$

Example 8

Find an annihilator operator for the functions:**(a)** $f(x) = e^{5x}$, **(b)** $g(x) = 4e^{2x} - 6xe^{2x}$

Solution

(a) Since $(D-5)e^{5x} = 5e^{5x} - 5e^{5x} = 0.$

Therefore, the annihilator operator of function f is given by $L = D-5$

We notice that in this case $\alpha = 5,\ n = 1$.

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MC210201002: AMIR SHAHZAD

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MTH401:Quiz NO.02

Quiz Start Time: 12:24 PM, 15 June 2021

Question # 4 of 10 (Start time: 12:26:57 PM, 15 June 2021)

Total Marks: 1

The input function $g(x)$ can have one of the forms: A constant function k; A polynomial function; _____.

Select the correct option

Reload Math Equations

- (III) Finite sums and products of these functions
- (I) An exponential function $e^{\alpha x}$
- (IV) All (I), (II) and (III).
- (II) The trignometric functions $\sin \beta x$, $\cos \beta x$

Click to Save Answer & Move to Next Question

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MC210201002: AMIR SHAHZAD

Time Left

MTH401:Quiz NO.02

Quiz Start Time: 12:24 PM, 15

Question # 2 of 10 (Start time: 12:24:48 PM, 15 June 2021)

Total

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

Reload Math E

- D
- SD
- D+1
- D+S

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The equation of free un-damped motion is _____.

elect the correct option

$\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$

$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$



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MTH401:Quiz NO.02

Question # 7 of 10 (Start time: 12:38:39 PM, 14 June 2021)

$Wronskian\ of\ 1\ and\ x^2,\ W(1,x^2) =$ _____

Select the correct option

- None of these
- 0
- x
- 2x

Click to S

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MTH401:Quiz NO.02 Quiz Start Time: 12:31 PM, 14 June 202

Question # 6 of 10 (Start time: 12:38:11 PM, 14 June 2021) Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- $A\cos 4x + B\sin 4x$
- $(A + B)(\cos 4x + \sin 4x)$
- Ae^{4x}
- $Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question

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Question # 8 of 10 (Start time: 12:27:38 PM, 14 June 2021)

The differential operator D transforms a _____ function into another function.

Select the correct option

- differentiable
- continuous
- discontinuous
- integrable

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H401:Quiz NO.02 Quiz Start Time: 12:3

stion # 5 of 10 (Start time: 12:37:04 PM, 14 June 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

ect the correct option

Relo

- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $(Ax^2 + Bx + C)e^{5x}$
- $(Ax + B)e^{5x}$
- Ae^{5x}

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MTH401:Quiz NO.02

Question # 4 of 10 (Start time: 12:36:07 PM, 14 June 2021)

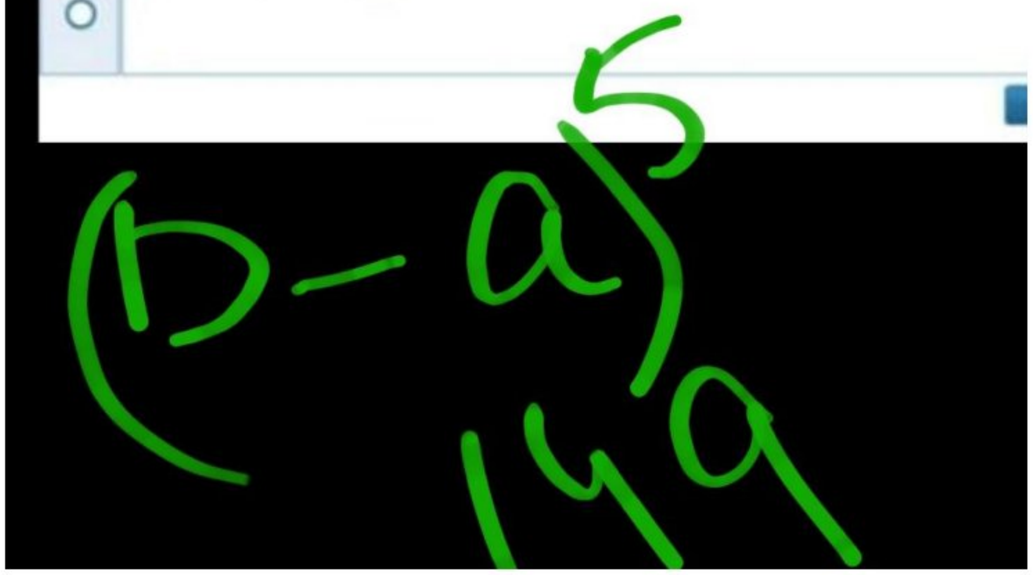
If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

- Its annihilator operator is (D-5)
- Its annihilator operator is $(D-5)^2$
- Its annihilator operator is $(D+5)^2$
- Its annihilator operator is (D+5)



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MTH401:Quiz NO.02 Quiz Start Time: 12:31 PM, 1

Question # 2 of 10 (Start time: 12:33:46 PM, 14 June 2021) T

Wronskian

$$W(x, 2x)$$

=_____.

Select the correct option

Reload Math

- 1
- None of these
- 0
- -1

Click to Save Answer & Move to Next

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MTH401:Quiz NO.02

Question # 3 of 10 (Start time: 12:34:58 PM, 14 June 2021)

Wronskian of the functions, $y_1(x) = 1$, $y_2(x) = \sin x$, $y_3(x) = \cos x$, is _____.

Select the correct option

- 0
- 1
- 2
- −1

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Question # 10 of 10 (Start time: 12:28:59 PM, 14 June 2021)

The displacement measured below the equilibrium position are _____.

Select the correct option

negative

positive

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Question # 5 of 10 (Start time: 12:24:10 PM, 14 June 2021) Total Marks:

We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with _____

Select the correct option

- Newton's 3rd law
- Hook's law

Click to Save Answer & Move to Next Question

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Question # 3 of 10 (Start time: 12:21:28 PM, 14 June 2021)

For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

Select the correct option

- Revolution
- Period
- Amplitude
- Frequency

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MTH401:Quiz NO.02 Quiz Start Time: 12:24 P

Question # 4 of 10 (Start time: 12:26:57 PM, 15 June 2021)

The input function $g(x)$ can have one of the forms: A constant function k; A polynomial function; _____

Select the correct option

Reload

- (III) Finite sums and products of these functions
- (I) An exponential function $e^{\alpha x}$
- (IV) All (I), (II) and (III).
- (II) The trignometric functions $\sin \beta x$, $\cos \beta x$

Click to Save Answer & Move to Next Q

1:57 pm

19 Undetermined Coefficients(Annihilator Operator Approach)

The method of undetermined coefficients that utilizes the concept of annihilator operator approach is also limited to non-homogeneous linear differential equations

- That have constant coefficients, and
- Where the function $g(x)$ has a specific form.

The form of $g(x)$ **:**The input function $g(x)$ has to have one of the following forms:

- A constant function k .
- A polynomial function
- An exponential function e^{X}
- The trigonometric functions $\sin(\beta x)$, $\cos(\beta x)$
- Finite sums and products of these functions.

Otherwise, we cannot apply the method of undetermined coefficients.

19.1 Solution Method

Consider the following non-homogeneous linear differential equation with constant coefficients of order n

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MTH401:Quiz NO.02 Quiz Start Time: 12:31 PM, 14 June 20

Question # 5 of 10 (Start time: 12:37:04 PM, 14 June 2021) Total Marks

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equation

- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $(Ax^2 + Bx + C)e^{5x}$
- $(Ax + B)e^{5x}$
- Ae^{5x}

Click to Save Answer & Move to Next Question

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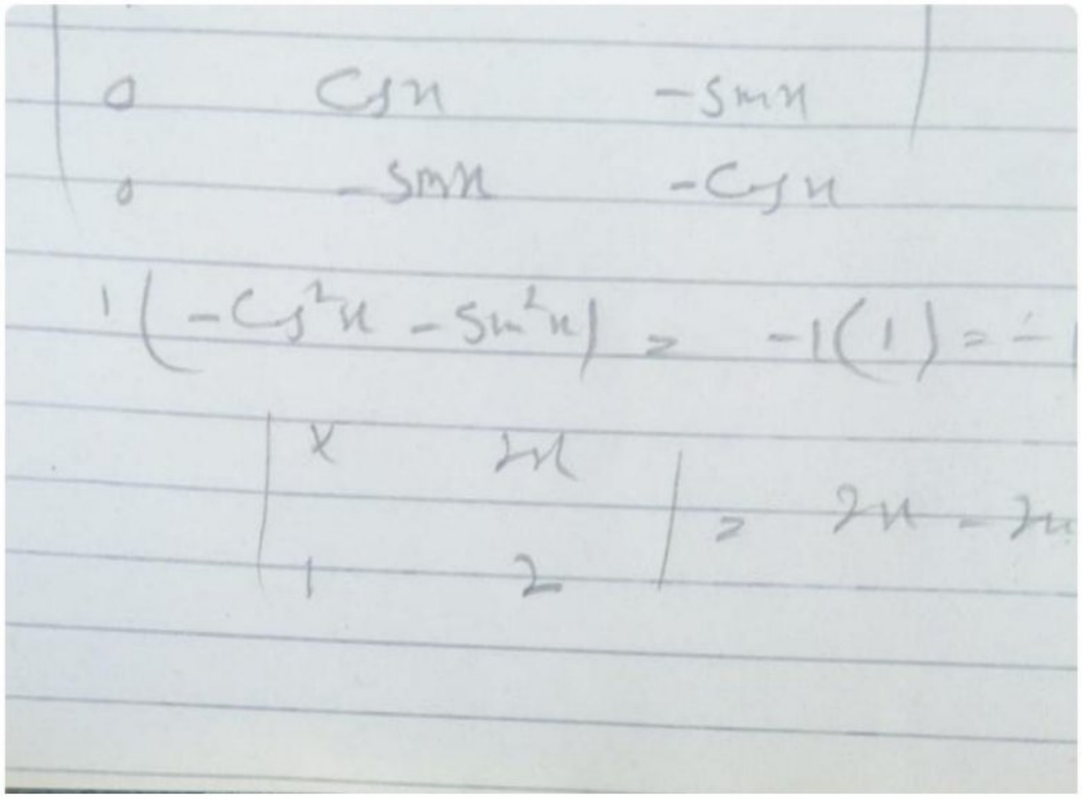
MTH401:Quiz NO.02

Question # 3 of 10 (Start time: 12:34:58 PM, 14 June 2021)

Wronskian of the functions, $y_1(x) = 1,\ y_2(x) = \sin x,\ y_3(x) = \cos x$, is _____.

Select the correct option

- 0
- 1
- 2
- −1 ✓



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MTH401:Quiz NO.02

Question # 3 of 10 (Start time: 12:21:28 PM, 14 June 2021)

For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

Select the correct option

- Revolution
- Period
- Amplitude
- Frequency

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MTH401:Quiz NO.02 Quiz Start Time: 12:19 PM, 14 June 202

Question # 4 of 10 (Start time: 12:23:17 PM, 14 June 2021) Total Marks:

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

Select the correct option

Reload Math Equations

- $m^3 + 2m^2 - 3 = 0$
- $m^3 + 2m^2 = 0$
- $m^3 - 3 = 0$
- $(m^3 + 2m^2 - 3)\, e^{mx} = 0$

Click to Save Answer & Move to Next Question

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MTH401:Quiz NO.02 Quiz Start Time: 12:19 PM, 14 June 202

Question # 7 of 10 (Start time: 12:26:04 PM, 14 June 2021) Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- $(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
- $A \cos 4x + B \sin 4x$
- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$

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Question # 10 of 10 (Start time: 10:45:38 PM, 15 June 2021)

Total Ma

The auxiliary equation of the differential equation, $3y''' + 5y'' + 10y' - 4y = 0$, is _____.

Select the correct option

Reload Math Equa

- $3m^3 + 5m^2 + 10m - 4 = 0$
- $3m^3 + 5m^2 + 10m$
- $\left(3m^3 + 5m^2 + 10m - 4\right)e^{mx} = 0$
- $3m^3 + 5m^2 - 4$

Click to Save Answer & Move to Next Question

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16.1.3 Case 3 (Complex Roots)

If the auxiliary equation has complex roots $\alpha \pm i\beta$ then, with $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where $\alpha > 0$ and $\beta > 0$ are real, the general solution of the differential equation is $y = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x}$

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Question # 6 of 10 (Start time: 10:41:46 PM, 15 June 2021)

A set of n functions

$$\{f_1(x), f_2(x), f_3(x), \ldots\ldots, f_n(x)\}$$

is _____ if at least one of them can be expressed as a linear combination of the remaining.

Select the correct option

- linear independent
- linear dependent





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38% 1:25 pm
linear 73 of 313

So that the functions are linearly dependent because $c_1 = -1$.

Hence, we conclude that:

- Any two functions $f_1(x)$ and $f_2(x)$ are linearly dependent on an interval I if and only if one is the constant multiple of the other.
- Any two functions are linearly independent when neither is a constant multiple of the other on an interval I.
- In general a set of n functions $\{f_1(x), f_2(x), \ldots, f_n(x)\}$ is linearly dependent if at least one of them can be expressed as a linear combination of the remaining.

Example 1

The functions

$$f_1(x) = \sin 2x, \qquad \forall x \in (-\infty, \infty)$$

$$f_2(x) = \sin x \cos x, \quad \forall x \in (-\infty, \infty)$$

If we choose $c_1 = \frac{1}{2}$ and $c_2 = -1$ then

$$c_1 \sin 2x + c_2 \sin x \cos x = \frac{1}{2}(2 \sin x \cos x) - \sin x \cos x = 0$$

Hence, the two functions $f_1(x)$ and $f_2(x)$ are linearly dependent.

Example 2

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linear 99 of 313

Hence Wronskian of y_1 and y_2 is either identically zero or is never zero on I.

In general

If $y_1, y_2, \ldots, y_n$ are n solutions, on an interval I, of the homogeneous nth order linear differential equation with constants coefficients

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

Then

Either $$W(y_1, y_2, \ldots, y_n) = 0, \quad \forall x \in I$$

or $$W(y_1, y_2, \ldots, y_n) \neq 0, \quad \forall x \in I$$

14.3 Linear Independence of Solutions

Suppose that

$$y_1, y_2, \ldots, y_n$$

are n solutions, on an interval I, of the homogeneous linear nth-order differential equation

$$\left(\quad\right)\frac{d^n y}{\quad} \quad \left(\quad\right)\frac{d^{n-1} y}{\quad} \quad \left(\quad\right)\frac{dy}{\quad} \quad \left(\quad\right)$$

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MTH401:Quiz NO.02

Question # 5 of 10 (Start time: 10:39:39 PM, 15 June 2021)

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

- 1
- 0
- $-3x$
- $3x$

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M401:Quiz No.02 Quiz Start Time: 10:31 I

estion # 1 of 10 (Start time: 10:31:43 PM, 15 June 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.

If the input function, $g(x) = (9x - 2)e^{3x}$, then the assumed particular solution y_p could be _____.

ect the correct option

Rela

- $Ae^{3x}\cos 5x + Be^{3x}\sin 5x$
- $(Ax^2 + Bx + C)e^{3x}$
- $(Ax + B)e^{3x}$
- Ae^{3x}

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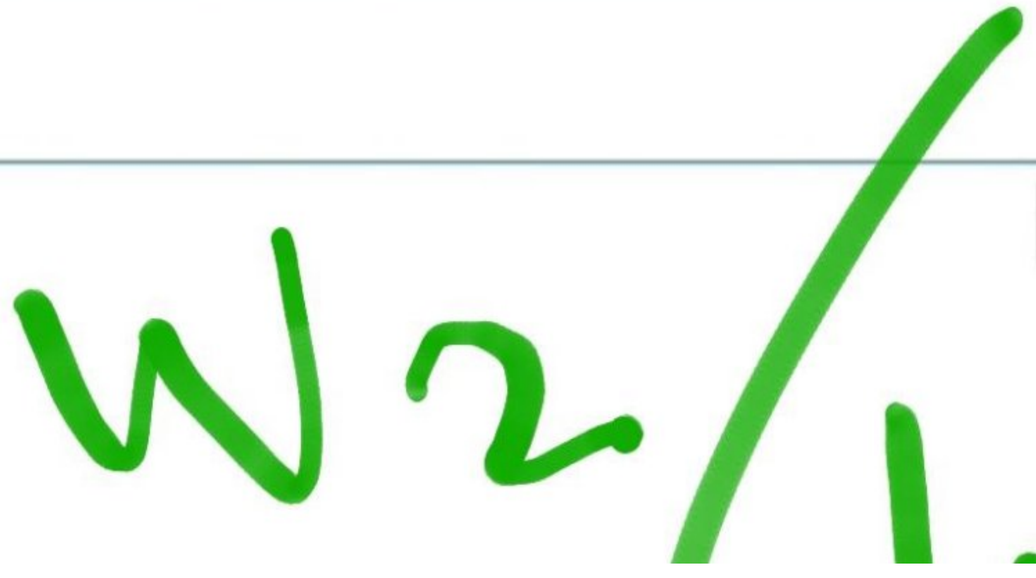
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uestion # 6 of 10 (Start time: 12:29:14 PM, 15 June 2021)

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 4$, $W_2 = -\frac{1}{5}$, then $u'_2 =$ _____.

elect the correct option

- $-\frac{1}{5}$
- 4
- $-\frac{1}{20}$
- -20



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MC210201002: AMIR SHAHZAD

Time Left 89 sec(s)

MTH401:Quiz NO.02

Quiz Start Time: 12:24 PM, 15 June 202

Question # 1 of 10 (Start time: 12:24:07 PM, 15 June 2021)

Total Marks:

Wronskian

$$W(1, e^x)$$

=_____.

Select the correct option

Reload Math Equations

1

0

-1

e^x

Click to Save Answer & Move to Next Question

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Question # 10 of 10 (Start time: 12:34:33 PM, 15 June 2021)

The derivatives ${u_1}', {u_2}', \ldots, {u_n}'$ of the unknown functions $u_1, u_2, \ldots, u_n$ are find by using

Select the correct option

- $u_k' = \frac{W}{W_k}, \quad k = 1, 2, \ldots, n$
- $u_k' = \frac{W_k}{W}, \quad k = 1, 2, \ldots, n$

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Question # 10 of 10 (Start time: 12:34:33 PM, 15 June 2021)

The derivatives $u_1', u_2', \ldots, u_n'$ of the unknown functions $u_1, u_2, \ldots, u_n$ are find by using

Select the correct option

- $u_k' = \frac{W}{W_k}, \quad k = 1, 2, \ldots, n$

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MTH401:Grand Quiz

Question # 6 of 30 (Start time: 03:30:01 PM, 03 January 2021)

The differential equation of orthogonal trajectory to the family of curves

$$x^2 + y^2 = C^2$$

is _____.

Select the correct option

- $\frac{dy}{dx} = \frac{x}{y}$
- $\frac{dy}{dx} = \frac{y}{x}$
- $\frac{dy}{dx} = -\frac{x}{y}$
- $\frac{dy}{dx} = -\frac{y}{x}$

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:10200757: ABBAS RAZA

Time Left 89 sec(s)

401:Quiz NO.02

Quiz Start Time: 10:11 AM, 17 June 2021

tion # 10 of 10 (Start time: 10:19:28 AM, 17 June 2021)

Total Marks: 1

The root(s) of the auxiliary equation, $am^2 + bm + c = 0$, for the differential equation, $ay'' + by' + cy = 0$, is (are) _____.

t the correct option

Reload Math Equations

(IV) Both (I) and (II).

(III) $m_3 = \dfrac{b + \sqrt{b^2 + 4ac}}{2a}$

(II) $m_2 = \dfrac{-b - \sqrt{b^2 - 4ac}}{2a}$

(I) $m_1 = \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}$

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Question # 9 of 10 (Start time: 10:18:49 AM, 17 June 2021)

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

- $D + 5$
- D^2
- D
- $D + 1$

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uestion # 7 of 10 (Start time: 10:16:55 AM, 17 June 2021) Total M

The velocity of a particle moving with simple harmonic motion is ---------- at the mean position.

lect the correct option

- None of these
- Maximum
- Zero
- Minimum

Click to Save Answer & Move to Next Question

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Question # 6 of 10 (Start time: 10:16:13 AM, 17 June 2021) Total Marks: 1

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____.

Select the correct option

- Hook's law
- Newton's third law
- Newton's first law
- Newton's second law

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ATH401:Quiz NO.02 Quiz Start Time: 10:11 AM, 17 June 2021

Question # 5 of 10 (Start time: 10:15:28 AM, 17 June 2021) Total Marks: 1

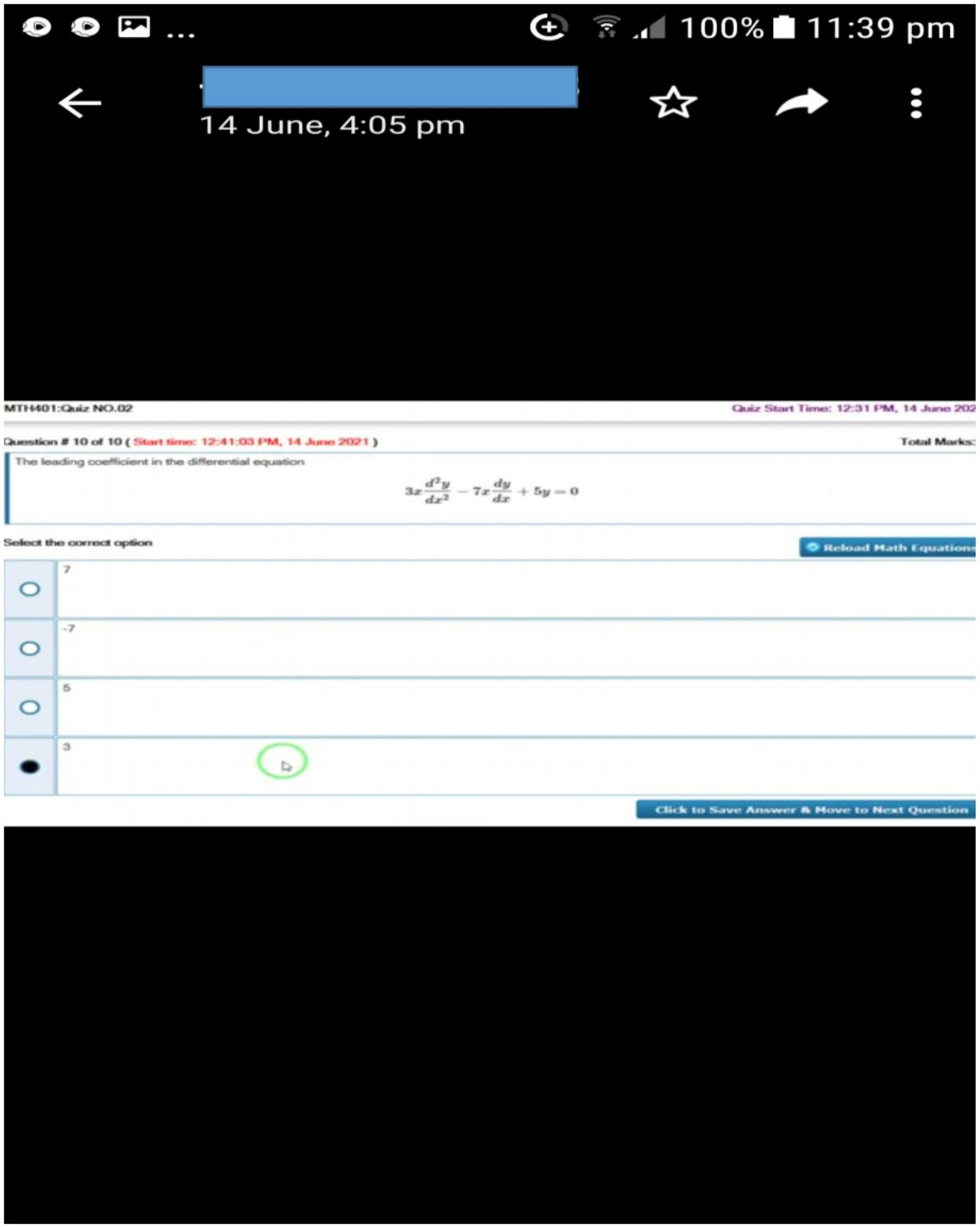
The hook's law states that the force F is proportional to the _____.

Select the correct option

- Weight
- Length
- None of these
- Elongation

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MTH401:Quiz NO.02 Quiz Start Time: 12:31 PM, 14 June 202

Question # 6 of 10 (Start time: 12:38:11 PM, 14 June 2021) Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- $A\cos 4x + B\sin 4x$
- $(A + B)(\cos 4x + \sin 4x)$
- Ae^{4x}
- $Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question

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MTH401:Quiz NO.02

Quiz Start Time: 12:31 PM, 14 June 202

Question # 5 of 10 (Start time: 12:37:04 PM, 14 June 2021)

Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $(Ax^2 + Bx + C)e^{5x}$

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MTH401:Quiz NO.02 Quiz Start Time: 12:31 PM, 14 June 202

Question # 3 of 10 (Start time: 12:34:58 PM, 14 June 2021) Total Marks:

Wronskian of the functions, $y_1(x) = 1,\ y_2(x) = \sin x,\ y_3(x) = \cos x$, is _____.

Select the correct option

Reload Math Equations

- ○ 0
- ○ 1
- ○ 2
- ● -1

Click to Save Answer & Move to Next Question

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ITH401:Quiz NO.02

Quiz Start Time

uestion # 2 of 10 (Start time: 12:33:46 PM, 14 June 2021)

Wronskian

$$W(x, 2x)$$

=_____.

elect the correct option

- 1
- None of these
- 0
- -1

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Question # 10 of 10 (Start time: 12:28:59 PM, 14 June 2021) Total Marks:

The displacement measured below the equilibrium position are _____.

Select the correct option

- negative
- positive

Click to Save Answer & Move to Next Question

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MTH401:Quiz NO.02

Question # 8 of 10 (Start time: 12:27:38 PM, 14 June 2021)

The differential operator D transforms a _____ function into another function

Select the correct option

- (●) differentiable
- ( ) continuous
- ( ) discontinuous

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[redacted]

MTH401:Quiz NO.02

Quiz Start Time: 12:19 PM, 14 June 202

Question # 7 of 10 (Start time: 12:26:04 PM, 14 June 2021)

Total Marks:

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- $(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
- $A \cos 4x + B \sin 4x$

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[redacted]

MTH401:Quiz NO.02

Quiz Start Time: 12:19 PM, 14 June 202

Question # 6 of 10 (Start time: 12:25:00 PM, 14 June 2021)

Total Marks:

The differential equation

$$4x^2\frac{d^2y}{dx^2} + 12x\frac{dy}{dx} + 3y = 0$$

subject to the condition

$$y(4) = \frac{1}{8}$$

and

Select the correct option

Reload Math Equations

- None of these

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[redacted]

Question # 6 of 10 (Start time: 12:25:00 PM, 14 June 2021) Total Marks:

subject to the condition

$$y(4) = \frac{1}{8}$$

and

$$y'(4) = -\frac{3}{64}$$

is known as_____.

Select the correct option

Reload Math Equations

- None of these
- Initial value problem

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Question # 5 of 10 (Start time: 12:24:10 PM, 14 June 2021) Total Marks:

We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with _____.

Select the correct option

- Newton's 3rd law
- Hook's law

Click to Save Answer & Move to Next Question

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Question # 4 of 10 (Start time: 12:23:17 PM, 14 June 2021) Total Marks:

The auxiliary equation of the differential equation, $y''' + 2y'' - 3y = 0$, is _____.

Select the correct option

Reload Math Equations

- $m^3 + 2m^2 - 3 = 0$
- $m^3 + 2m^2 = 0$
- $m^3 - 3 = 0$
- $\left(m^3 + 2m^2 - 3\right) e^{mx} = 0$

Click to Save Answer & Move to Next Question

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MTH401:Quiz NO.02

Question # 3 of 10 (Start time: 12:21:28 PM, 14 June 2021)

For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

Select the correct option

- Revolution
- Period
- Amplitude

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MTH401:Quiz NO.02 Quiz Start Time: 12:19 PM, 14 June 202

Question # 2 of 10 (Start time: 12:20:07 PM, 14 June 2021) Total Marks:

If the auxiliary equation, $am^2 + bm + c = 0$, has two unequal real roots m_1 and m_2 then the solutions of the differential equation, $ay'' + by' + cy = 0$, could be _____.

Select the correct option

Reload Math Equations

- (I) $y = c\,e^{mx}$
- (IV) Both (II) and (III).
- (III) $y_2 = e^{m_2 x}$
- (II) $y_1 = e^{m_1 x}$

Click to Save Answer & Move to Next Question

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The auxiliary equation of the differential equation, $3y''' + 5y'' + 10y' - 4y = 0$, is _____.

Select the correct option

- $3m^3 + 5m^2 - 4$
- $3m^3 + 5m^2 + 10m - 4 = 0$
- $\left(3m^3 + 5m^2 + 10m - 4\right) e^{mx} = 0$
- $3m^3 + 5m^2 + 10m$

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Question # 9 of 10 (Start time: 11:43:20 PM, 15 June 2021)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) =$ constant (say 1), then the assumed particular solution y_p could be _____.

Select the correct option

Reload

- ● A
- ○ $Ax^2 + Bx + C$
- ○ $Ax + B$

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https://quiz.vu.edu.pk/QuizQuestion.aspx?

BC190201018: ANAM AMIR

MTH401:Quiz NO.02

Question # 8 of 10 (Start time: 11:42:51 PM, 15 June 2021)

The hook's law states that the force F is proportional to the _____.

Select the correct option

- Elongation
- None of these

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Question # 7 of 10 (Start time: 11:42:07 PM, 15 June 2021)

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

- (ii) y = f(x) defines a sufficiently differentiable function.
- (i) L is a linear differential operator with constant coefficients.
- (iii) The function f is such that L(y)=0.
- all

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BC190201018: ANAM AMIR

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MTH401:Quiz NO.02

Quiz Start Time: 11:38 PM, 15 June 2021

Question # 6 of 10 (Start time: 11:41:20 PM, 15 June 2021)

Total Marks: 1

If the auxiliary equation $m^3 + m = 0$ has roots $m = 0, m = \pm i$ then the complementary function is

Select the correct option

Reload Math Equations

- $y_c = c_1 \cos x + c_2 \sin x$
- $y_c = c_1 + c_2 \cos x + c_3 \sin x$

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https://quiz.vu.edu.pk/QuizQuestion.aspx?ver=86e43fff-9bd2-4bab-a6f6-cde3414f18d8

If 3,3 are real roots of a differential equation, then the general solution is_____.

Select the correct option

Reload Math Equations

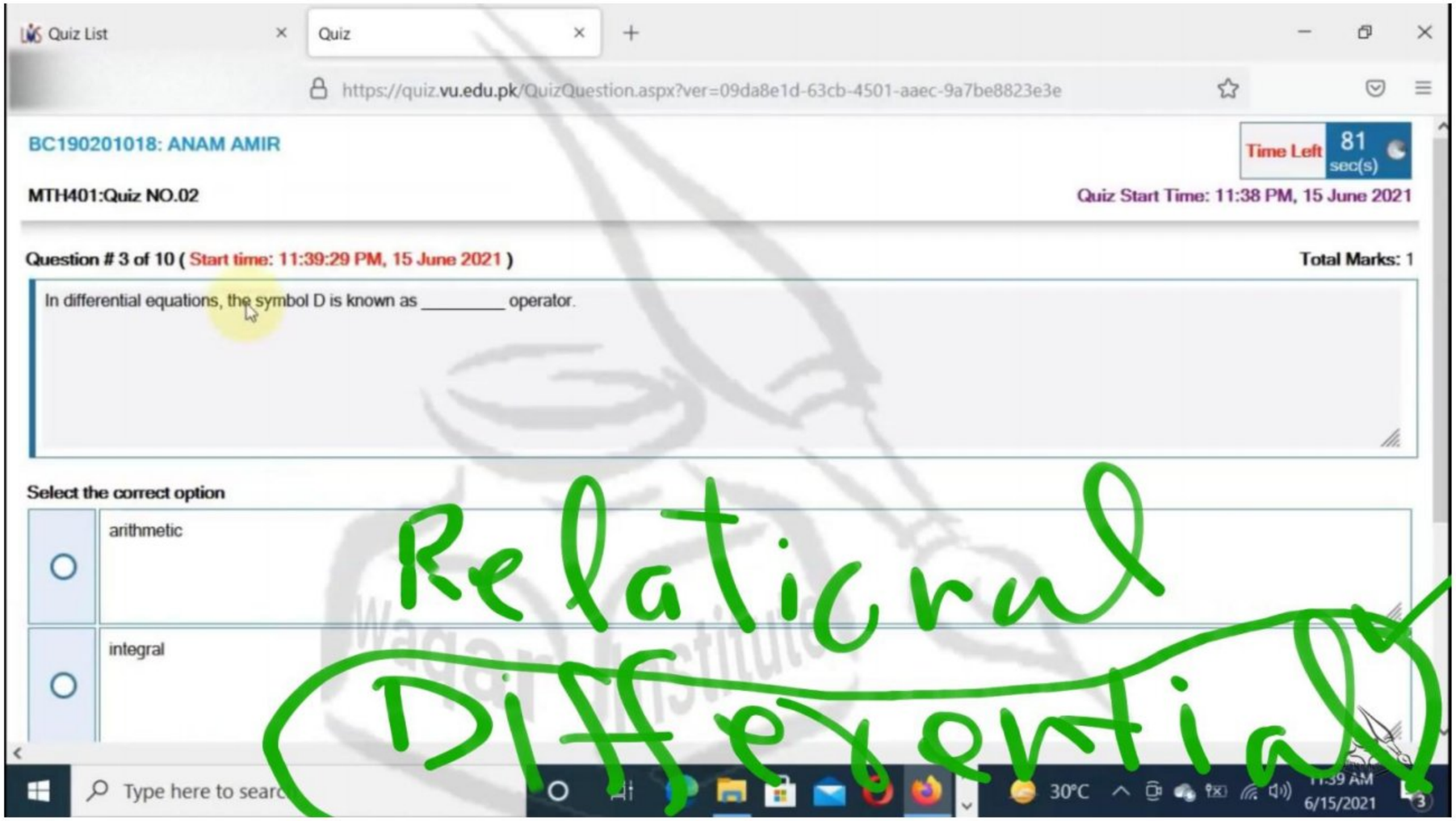
- $y = c_1 e^{3x} + c_2 e^{3x}$
- $y = c_1 e^{3x}$
- $y = (c_1 + c_2 x) e^{3x}$ ✓
- $y = c_1 e^{3x} - c_2 e^{3x}$

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BC190201018: ANAM AMIR

MTH401:Quiz NO.02

Question # 4 of 10 (Start time: 11:39:44 PM, 15 June 2021)

The equation of free un-damped motion is _____.

Select the correct option

- $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
- $\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$

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Question # 1 of 10 (Start time: 11:38:21 PM, 15 June 2021) **Total Marks: 1**

Following is (are) the form(s) of the general solution of the differential equation, $ay'' + by' + cy = 0$, corresponding to the case(s): _____.

Select the correct option

Reload Math Equations

- (●) (IV) All (I), (II) and (III).
- ( ) (III) m_1 and m_2 conjugate complex numbers i.e. $(b^2 - 4ac < 0)$
- ( ) (II) m_1 and m_2 real and equal i.e. $(b^2 - 4ac = 0)$
- ( ) (I) m_1 and m_2 real and distinct i.e. $(b^2 - 4ac > 0)$

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Time Left 82 sec(s)

MTH401:Quiz NO.02

Quiz Start Time: 11:38 PM, 15 June 2021

Question # 2 of 10 (Start time: 11:39:02 PM, 15 June 2021)

Total Marks: 1

$$4x^2\frac{d^2y}{dx^2} + 12x\frac{dy}{dx} + 3y = 0$$

subject to the condition

$$y(4) = \frac{1}{8}$$

and

$$y'(4) = -\frac{3}{64}$$

10 seconds

Select the correct option

Reload Math Equations

- Boundary value problem
- None of these

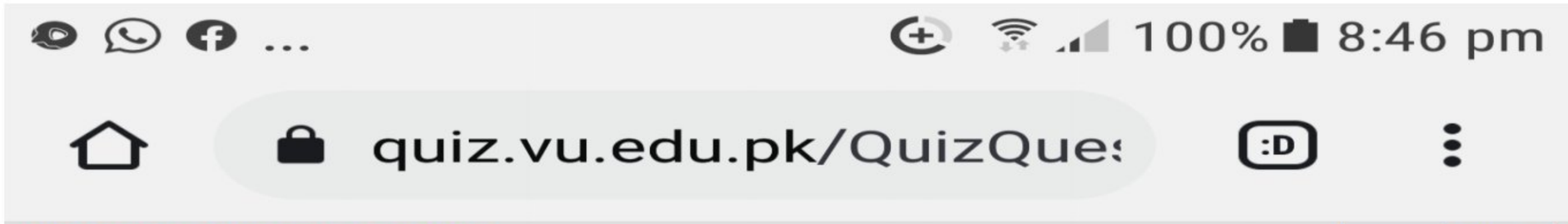






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MC210203146: AMINA SAMI-UR-REHMAN

Time Left 35 sec(s)

MTH501:Quiz No 2

Quiz Start Time: 08:42 PM, 16 June 2021

Question # 10 of 10 (Start time: 08:45:12 PM, 16 June 2021)

Total Marks: 1

Let P_n be the set of real valued function of the form $P(x) = a_0 + a_1x + a_2x^2 + \ldots + a_nx^n$ wheren is a non-negative integer and $a_0, a_1, \ldots, a_n$ are real numbers, then P_n is a ---------- $F(-\infty, + + \infty)$.

Select the correct option

Reload Math Equations

- Subspace
- None of the above
- Space
- Vector Space

Click to Save Answer & Move to Next Question

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